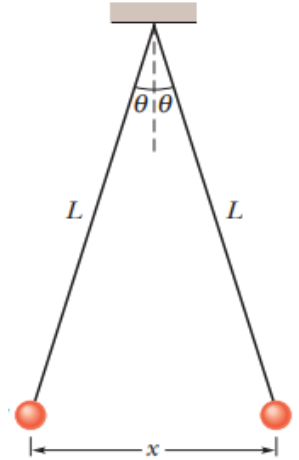


**PHY 109**  
**Assignment: Total Marks - 20**  
**Last date of submission: 06.04.2026**  
**Sec-03**

*Each question contains 5 marks*

**Q1.** In Fig. 1, two tiny conducting balls having identical mass  $m$ , but different charges, one is neutral ( $q = 0$ ) and another has  $q_0$  charge hang from nonconducting threads of length  $L$ . After they contact each other, the balls gain an equilibrium state. Assume that  $\theta$  is so small that  $\tan \theta$  can be replaced by its approximate equal,  $\sin \theta$ .

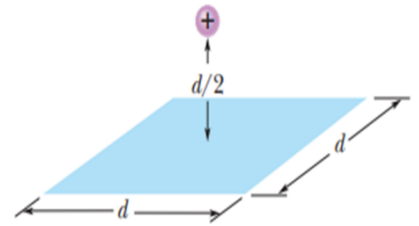


*Fig. 1*

(a) Show that,  $x = \left( \frac{q_0^2 L}{8\pi\epsilon_0 m g} \right)^{\frac{1}{3}}$  gives the equilibrium separation  $x$  of the balls.

(b) If  $L = 120 \text{ cm}$ ,  $m = 10\text{g}$ , and  $x = 5.0 \text{ cm}$ ,  
what is  $|q_0|$ ?

**Q2.** In Fig. 2, a proton is a distance  $\frac{d}{2}$  directly above the center of a square of the side  $d$ . What is the magnitude of the electric flux through the square? (Hint: Think of the square as one face of a cube with an edge  $d$ .)



*Fig. 2*

**Q3.** A particle of charge  $+q$  is placed at one corner of a Gaussian cube. What multiple of  $\frac{q_{enc}}{\epsilon_0}$  gives the flux through (a) each cube face forming that corner and (b) each of the other cube faces?

**Q4.** Charge is distributed uniformly throughout the volume of an infinitely long solid cylinder of radius  $R$ . (a) Show that, at a distance  $r < R$  from the cylinder axis,

$$E = \frac{\rho r}{2\epsilon_0}$$

where  $\rho$  is the volume charge density.

(b) Write an expression for  $E$  when  $r > R$ .